

A Computer-Assisted Proof of Erdős Problem 848

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Computer-assisted research and verification with OpenAI Codex

11 August 2026

Abstract

We prove that, for every positive integer N , the largest $A \subseteq \{1, \dots, N\}$ for which $ab + 1$ is nonsquarefree for all $a, b \in A$, including $a = b$, has cardinality $\lfloor (N + 18)/25 \rfloor$. The proof combines an elementary lower construction, an independently checked exact coloring certificate through 100000006, exact-rational structural interval certificates through 2640000000000000000, and an independently audited explicit high-range theorem. We state the certificate semantics and trusted execution boundary explicitly.

1 Statement and sharpness

For positive N , call $A \subseteq [N] = \{1, \dots, N\}$ *admissible* when $ab + 1$ is nonsquarefree for all ordered pairs $a, b \in A$, including the diagonal. Let $f(N)$ be the maximum cardinality of such a set and set

$$B_7(N) = \left\lfloor \frac{N + 18}{25} \right\rfloor.$$

Theorem 1. *For every positive integer N ,*

$$f(N) = B_7(N).$$

The theorem is the exact source-audited statement; it makes no uniqueness claim. The residue class

$$A_7(N) = \{a \in [N] : a \equiv 7 \pmod{25}\}$$

is admissible, since $ab + 1 \equiv 7^2 + 1 = 50 \equiv 0 \pmod{25}$ for $a, b \equiv 7 \pmod{25}$. Writing $N = 25q + r$, $0 \leq r < 25$, shows $|A_7(N)| = q + \mathbf{1}_{r \geq 7} = B_7(N)$. Hence $f(N) \geq B_7(N)$.

2 Graph reduction

Let

$$D_N = \{a \in [N] : a^2 + 1 \text{ is nonsquarefree}\}.$$

Form G_N on D_N , joining distinct a, b when $ab + 1$ is nonsquarefree. The diagonal condition places every admissible element in D_N , and the remaining conditions say exactly that the set is a clique. Thus $f(N) = \omega(G_N)$. A proper coloring of G_N has squarefree products plus one within each color class, so

$$\chi(G_N) \leq B_7(N) \implies f(N) \leq B_7(N).$$

3 Exact finite theorem

Put $e_i = 25i + 6$. Since $B_7(N) = i$ for $25i - 18 \leq N \leq 25i + 6$, an i -coloring at e_i restricts to every prefix in that block. There are no diagonal-eligible vertices below 7.

3.1 Diagonal completeness

If $p^2 \mid a^2 + 1$, then $p \neq 2$, $p \equiv 1 \pmod{4}$, and $p^2 \leq a^2 + 1 < (a + 1)^2$, so $p \leq a$. For every prime $p \leq 100000006$ with $p \equiv 1 \pmod{4}$, the independent checker obtains both roots of -1 modulo p by Tonelli–Shanks and uniquely Hensel-lifts them modulo p^2 . Marking the two progressions therefore reconstructs exactly all diagonal vertices. The count is 10515898.

3.2 Endpoint induction

The base stream supplies all 4000 endpoint states through 100006. At endpoint e_i the checker colors every diagonal vertex through e_i with a color below i and checks every same-color pair. The compact stream supplies 3996000 later transitions. Each transition introduces the two principal anchors, performs no swap or one old-bin swap of an 18-anchor, and places the exact newly exposed outsiders. Every changed bin is checked; unchanged bins retain the induction invariant. Hence every endpoint through $e_{4000000} = 100000006$ has the required coloring.

The exhaustive replay checked 18049789 base pair occurrences. It also checked all 14458371 compact affected-bin pair occurrences, together with 1379312 swaps and 2513387 placements.

3.3 Exact factor leaves

Every accepted pair value is factored by small-prime division and deterministic Brent splitting. The checker verifies the exact factor product and absence of repeated factors and exports every terminal leaf used by an accepted computation. Miller–Rabin only proposes terminal leaves; all such leaves are independently certified below.

There are 11108162 distinct leaves, with maximum $m = 9999932500113877$ and $L = \lfloor \sqrt{m} \rfloor = 99999662$. An exact sieve proves the 2328675 leaves at most L prime. Let M be the primorial through L and P the product of the 8779487 larger leaves. Exact GMP arithmetic gives $\gcd(P, M) = 1$. A composite large leaf would have a prime divisor at most L , contradicting this gcd. Thus all leaves are prime, and every accepted pair value is squarefree.

The canonical leaf receipt has SHA-256 `4fc75b0ed263df81c07c5997c915511351ea61fff085d3ac95159c524ca9aad6`. The endpoint-401 mutation temporarily puts 18 with 382, for which $18 \cdot 382 + 1 = 13 \cdot 23^2$, and repairs the later state. The exhaustive checker rejects it at endpoint 401; sampled checking does not. Therefore

$$f(N) \leq B_7(N) \quad (1 \leq N \leq 100000006).$$

4 Structural reduction

Fix a positive integer N and put

$$A_{18}(N) = \{a \in [N] : a \equiv 18 \pmod{25}\}, \quad O_N = D_N \setminus (A_7(N) \cup A_{18}(N)).$$

The roots of -1 modulo 25 are 7 and 18; call the elements of O_N outsiders. For $z \in O_N$, let

$$w(z) = \min\{p : p \text{ prime and } p^2 \mid z^2 + 1\}.$$

An outsider has witness 13 or at least 17. For a possible least witness p , write $W_{p,N} = \{z \in O_N : w(z) = p\}$ and $U_{>p,N} = \bigcup_{q>p} W_{q,N}$, with q prime. For $z \in W_{p,N}$, put

$$L_{z,N} = \{x \in A_7(N) : zx + 1 \text{ is nonsquarefree}\}, \quad R_{z,N} = \{y \in A_{18}(N) : zy + 1 \text{ is nonsquarefree}\}.$$

Join $x \in L_{z,N}$ and $y \in R_{z,N}$ when $xy + 1$ is squarefree, obtaining a bipartite graph $H_{z,N}$. Define

$$s_N(z) = |L_{z,N}| + |R_{z,N}| - \nu(H_{z,N}), \quad c_{>p,N}(z) = |\{u \in U_{>p,N} : zu + 1 \text{ is nonsquarefree}\}|,$$

and $g_{p,N}(z) = s_N(z) + c_{>p,N}(z)$.

Lemma 1 (Structural rank). *If $A \subseteq [N]$ is admissible and has globally least outsider witness p , then*

$$|A| \leq \max_{1 \leq t \leq |W_{p,N}|} (t + g_{p,N}^\downarrow(t)),$$

where the scores are ordered decreasingly.

Proof. Let $T = A \cap W_{p,N}$ and $t = |T|$. For $z \in T$, the principal part of A is an independent set in $H_{z,N}$, hence has size at most $s_N(z)$ by König's theorem. Every higher-witness outsider in A is counted by $c_{>p,N}(z)$. Thus $|A| \leq t + \min_{z \in T} g_{p,N}(z)$. The minimum on any t -element set is at most its t -th largest ambient score. $\square$

This gives three exhaustive branches: no outsider, least witness 13, and least witness at least 17. The no-outsider branch is the corresponding principal bipartite independent-set bound.

5 Exact structural interval certificates

Each certificate row is a closed interval $[A, B]$. Positive census, collision, and Pell terms are evaluated at a uniform upper endpoint; subtractive gains are evaluated at their weakest endpoint; and translation remainders are maximized over the complete 25-cycle. The resulting exact integer or rational is compared with $B_7(A)$. Since B_7 is nondecreasing, an accepted row proves every integer in its interval.

The independently audited envelope inputs are exact residue censuses and CRT counts, the $p = 17$ orientation cap, the $p \geq 29$ domination bound, the no-outsider two-degree bound, the rank lemma above, and a generalized-Pell orbit bound. For the latter, solutions are grouped into ideals of norm dividing the fixed discriminant; two generators of one ideal differ by a totally positive norm-one unit. Such a nontrivial unit has trace at least 3, so successive positive solutions grow by more than $(3 + \sqrt{5})/2 > 2$. Every prime-count call satisfies the hypothesis of Hanson's strict bound $\pi(x) < 1.25506x/\log x$; the computation replaces the logarithm by a weaker exact-rational lower bound.

We make explicit one correction to the wording of the pinned `P13_RANK_ENVELOPE.md`. For a translated pair $x, x + d \in [N]$, put $\Delta = d^2 - 4$. If $x(x + d) + 1 = m\ell^2$ with $\ell > Y$, then $x(x + d) + 1 \leq N^2 + 1 < 2N^2$, so $m < 2N^2/Y^2$. The ideal-orbit bound gives at most $\tau(|\Delta|)K(B)$ solutions for each fixed m , where $K(B) = 1 + \lceil \log_2(4B) \rceil$. Hence the absolute count at the actual N is at most

$$\frac{2\tau(|\Delta|)K(B)N^2}{Y^2}.$$

After division by $M = N/25$, this is

$$\frac{50\tau(|\Delta|)K(B)N}{Y^2} \leq \frac{50\tau(|\Delta|)K(B)B}{Y^2},$$

exactly the term used by the verifier. The upstream prose instead passed from an absolute B^2 bound directly to the last display, which does not follow; the actual- N derivation above is the required repair and changes no certificate value.

The exact certified cover is:

Range	Method
$[10^8, 10^9]$	37 exact $p = 13$ rows and all other branches
$[10^9, 10^{12}]$	exact-rational short-shift rows, all branches
$[10^{12}, 2640000000000000000]$	1255 exact-rational blocks

The detailed formulas, endpoint choices, and row-soundness derivations are in the pinned ART-005 notes P13_INTERVAL_100M_1B.md, P17_EXACT_HYBRID.md, P13_DIRECT_RESIDUAL.md, LOW_RANGE_DIRECT_ENVELOPES.md, P13_RANK_ENVELOPE.md, and P17PLUS_AND_BASE_ENVELOPES.md; their assembly is ALL_N_THEOREM.md, at immutable revision 1afd7c722cae5ee7dd0fd1fde64427537394f749. The named paths resolve at <https://github.com/ipitchford/erdos-848-all-n/tree/1afd7c722cae5ee7dd0fd1fde64427537394f749/proofs> and are materialized locally by REPRODUCE.md. A referee-oriented extraction of the load-bearing claims is <https://github.com/dannyward630/erdos-848-proof/blob/main/proof/structural-certificate-appendix.md>.

The lower- $p = 13$ transcript contains 1439686 exact factorizations. An independent parser checks every product back to $z^2 + 1$. An exact sieve through 54392469 and a separate primorial gcd prove all 1357591 distinct claimed factors prime, so the selected collision states are exact. The original shared Miller–Rabin routine is not a premise.

The public release’s 19-stage full replay regenerated the censuses, factor states, masks, exact-rational rows, mutations, sanitizers, and closed coverage in a fresh extraction. It passed in 5523.791 seconds. The range-ledger digest is b28760bca88b3f4a356f5212f5aa3711df00ee527606058a9aefc193e715ebe1. Together with the structural lemma and audited envelopes, this proves

$$f(N) \leq B_7(N) \quad (100000000 \leq N \leq 2640000000000000000).$$

6 Explicit high range and conclusion

Sothanaphan’s independently audited explicit theorem proves the same upper bound for every $N \geq 2640000000000000000$. Its exact-rational numerical claims have digest 4726814d80fc63353a77c18bac8691bc917efd5ef1c7dcf10a14aed03674b215. For clarity, put

$$C_{\text{quad}} = 1 - \prod_{\substack{p \equiv 1 \pmod{4} \\ p \geq 13}} \left(1 - \frac{2}{p^2}\right).$$

The corrected form of the note’s misprinted Corollary 2 is the following: if $A \subseteq [N]$ is admissible, $N \geq 2640000000000000000$, and $A^* = A \setminus (A_7(N) \cup A_{18}(N))$, then

$$\frac{|A^*|}{N} \leq \frac{23}{25} C_{\text{quad}} + 0.001973;$$

if A^* consists only of odd elements, then

$$\frac{|A^*|}{N} \leq \frac{23}{50} C_{\text{quad}} + 0.001973.$$

Indeed, admissibility on (x, x) gives $\mu(x^2 + 1) = 0$ for every $x \in A^*$. Apply Proposition 2 of the note to the 23 nonprincipal classes modulo 25, or to their 23 odd lifts modulo 50, writing $q = 25$ or 50, respectively, and taking $R = \lceil (259/200)N^{2/3} \rceil$. The main terms sum to at most $23(N/q + 1)C_{\text{quad}}$. The pinned exact monotonicity audit bounds the remaining normalized error per class by 8.577×10^{-5} at the threshold and thereafter, while $23C_{\text{quad}}/N < 3 \times 10^{-18}$. Summing and rounding upward gives 0.001973. This proves the corrected corollary and identifies the diagonal step for which admissibility is required. Every invocation in Theorem 1 supplies that hypothesis.

The finite interval $[1, 100000006]$ overlaps the first structural interval $[100000000, 10^9]$. The remaining ranges share the endpoints 10^9 , 10^{12} , and 264000000000000000 . Their union is every positive integer. Hence $f(N) \leq B_7(N)$ for all positive N ; the residue class construction gives the reverse inequality and proves the theorem.

Decisive external references

- Denis Hanson, “On the Product of the Primes,” *Canadian Mathematical Bulletin* 15(1) (1972), 33–37, proves the strict prime-count estimate used above for every real $x > 1$. <https://doi.org/10.4153/CMB-1972-007-7>. The audited cached PDF is `sources/cache/hanson-1972-product-primes.pdf`, SHA-256 `ba350b2ce48e0ddb0751d8a60bcfe310683bc88df4fcf1b6a91029297688689c`.
- Nat Sothanaphan, *An Explicit Threshold in Erdős Problem #848*, 24 March 2026, proves the upper bound used above for every $N \geq 264000000000000000$. https://drive.usercontent.google.com/download?id=1ujhm4_WYpgRV_rdlrJXIIfHyvx16C0EKe&export=download&confirm=t. The audited cached PDF is `sources/cache/sothanaphan-2.64e17.pdf`, SHA-256 `8162113a571dc2283fc77de1cdf36e7abf424eeec952aa27cf82a4f44b3a796f`.

Exact retrieval and authentication commands are in `REPRODUCE.md`.

7 Assurance boundary

This is a computer-assisted proof using exact integer or rational arithmetic, authenticated certificate bytes, complete interval coverage, independent finite checking, batch primality certificates, and negative controls. Numerical evidence alone is not used as proof. Reproduction commands and decisive hashes are in `REPRODUCE.md`; complete run identities and receipts are in `certificates/`.

The research, implementation, drafting, and separated review lanes were carried out in a multi-agent OpenAI Codex workflow under Danny Ward’s direction. Here “independent” means a distinct implementation or internal review assignment, not external scholarly peer review. This public proof candidate is unrefereed; full provenance is in `PROVENANCE.md`.

The mathematical proof does not depend on the separate ART-006 Lean artifact. ART-006’s own publication state reports a closed kernel theorem with axiom closure drawn only from

`propext,      Classical.choice,      Quot.sound.`

Our source audit found that interface coherent and noncircular. This is strong upstream evidence, but not the independent clean rebuild required here. The repository’s stricter completion protocol keeps formal certification open until a sufficiently provisioned Windows host builds the exact 30636-module provider from source, performs trust-zero replay, and captures the live `#print axioms` and dependency audit.